Christoph Gerhardt | Dr.-Ing.
Candidate — Computer Vision & Deep Learning

Technische Universität Ilmenau – 98693 Ilmenau, Germany
christoph.gerhardt@tu-ilmenau.de • carhartt21.github.io • carhartt21
0009-0001-6821-7658 • Google Scholar Web of Science

Research Interests

Computer Vision, Semantic Segmentation, 360° Image Processing, Weather-Robust Perception, Data Augmentation, Outdoor Scene Understanding, Neural Network Architectures, Object Detection, Panoramic Imaging

Education

Technische Universität Ilmenau **Ilmenau, Germany**
Dr.-Ing. (PhD) in Computer Science *2019 – 2026 (exp.)*
Thesis: *Comprehensive Outdoor Scene Understanding Through Multi-Scale Neural Networks: From 360° Object Detection to Panoramic Segmentation and Weather-Robust Augmentation*
Supervisor: Prof. Dr. rer. nat. Wolfgang Broll
Research Group: Virtual Worlds and Digital Games (VWDG)

Technische Universität Ilmenau **Ilmenau, Germany**
M.Sc. Computer Engineering (Ingenieurinformatik), Grade: 1.5 (very good) *2014 – 2017*
Specialisation: Multimedia Information and Communication Systems
Thesis: *Selektive Verschlüsselung von Gesichtern in Videodaten unter Verwendung des H.264-Standards*

Technische Universität Ilmenau **Ilmenau, Germany**
B.Sc. Computer Engineering (Ingenieurinformatik), Grade: 1.8 (good) *2010 – 2014*
Thesis: *Entwicklung eines Verfahrens zur Verbesserung von Mikrofonklassifizierung basierend auf der Analyse von Umwelteinflüssen*

Gymnasium "Am Lindenberg" **Ilmenau, Germany**
Abitur, Grade: 1.5 *2009*

Professional Experience

Technische Universität Ilmenau **Ilmenau, Germany**
Research Associate (Wissenschaftlicher Mitarbeiter) *Feb 2019 – present*
Virtual Worlds and Digital Games Group (Prof. Broll)
- Research on neural-network-based semantic segmentation of outdoor scenes
- Development of weather-robust augmentation frameworks (AWARE / SWIFT / PRISM / PROVE)
- 360° panoramic image analysis for traffic sign detection and sky/cloud segmentation
- Co-supervision of student theses; teaching duties in VR/AR and computer vision
- Collaboration with NOVASIB GmbH on road inventory management (TT-SIB INFOSYS®)

Fraunhofer Institute for Digital Media Technology (IDMT) **Ilmenau, Germany**
Software Developer *Feb 2017 – Jan 2019*
Group: Virtual Acoustics
- Software development for spatial audio processing and rendering systems

Fraunhofer IDMT **Ilmenau, Germany**
Student Research Assistant (Hilfswiss. Mitarbeiter) *Jul 2016 – Jan 2017*
Group: Media Distribution and Security

Internships

Fraunhofer IDMT **Ilmenau, Germany**
Research Intern *Sep 2013 – Jan 2014*

Group: Media Distribution and Security

SIOS Meßtechnik GmbH — **Ilmenau, Germany**
Intern — *Mar 2013*

GPM Anlagenbau und Industrieanlagen GmbH — **Mainz, Germany**
Intern — *Sep – Oct 2012*

Civic Service

Führungsunterstützungsbataillon 292 — **Dillingen a. d. Donau, Germany**
Voluntary Military Service (Freiwilliger Wehrdienst) — *2009 – 2010*

Publications

Peer-Reviewed — First Author (Dissertation-Related)

[1] **C. Gerhardt** and W. Broll, "SkyCloud360: Sky and Cloud Segmentation in Equirectangular Images," in *Proc. 10th ICIVC*, Chengdu, China, 2025, pp. 48–58. doi:10.1109/ICIVC66358.2025.11200399.

[2] **C. Gerhardt**, F. Weidner, and W. Broll, "SkyCloud: Neural Network-Based Sky and Cloud Segmentation from Natural Images," in *Proc. 8th ICIVC*, Dalian, China, 2023. doi:10.1109/ICIVC58118.2023.10270450.

[3] **C. Gerhardt**, F. Weidner, and W. Broll, "OUTSIDE: Multi-Scale Semantic Segmentation of Universal Outdoor Scenes," in *Proc. IEEE 23rd MMSP*, Tampere, Finland, 2021, pp. 1–6. doi:10.1109/MMSP53017.2021.9733652.

[4] **C. Gerhardt** and W. Broll, "Neural Network-Based Traffic Sign Recognition in 360° Images for Semi-Automatic Road Maintenance Inventory," in *Proc. IEEE 23rd ITSC*, 2020, art. no. 9294610. doi:10.1109/ITSC45102.

[5] **C. Gerhardt**, M. M. Salman, and W. Broll, "AWARE: Augmentation for Weather-Adverse Robustness Evaluation," *IEEE Access*, 2025 *(under review)*.

Peer-Reviewed — Co-Author

[6] S. Arévalo Arboleda *et al.* incl. **C. Gerhardt**, "Robot, Avatar, or Human: The Impact of Partner Representation and Task on the Communication Experience," *Proc. ACM CSCW*, vol. 9, art. CSCW472, 2025. doi:10.1145/3757653.

[7] E. Makled, **C. Gerhardt**, T. Schwandt, F. Weidner, and W. Broll, "Evaluating Behavioral Realism in AR and VR...," *The Visual Computer*, vol. 41, no. 14, pp. 11721–11733, 2025. doi:10.1007/s00371-025-03934-5.

[8] V. Mikhailova *et al.* incl. **C. Gerhardt**, "Addressing the Avatar in the Room...," in *ITAP/HCII 2025*, LNCS 15809, Springer, 2025. doi:10.1007/978-3-031-92707-2_21.

[9] V. Mikhailova *et al.* incl. **C. Gerhardt**, "Work-in-Progress: Older Adults' Experiences with an AR Communication System," in *Proc. ACM IMX*, Stockholm, Sweden, 2024. doi:10.1145/3639701.3663641.

[10] E. Makled, **C. Gerhardt**, T. Schwandt, F. Weidner, and W. Broll, "Investigating Behavioral Realism of Single-Point IK Animated Avatars...," in *Proc. Int. Conf. Cyberworlds*, Kofu, Japan, 2024, pp. 1–8. doi:10.1109/CW64301.2024.00010.

[11] F. Weidner *et al.* incl. **C. Gerhardt**, "Eyes on the Narrative: Exploring the Impact of Visual Realism and Audio Presentation on Gaze Behavior in AR Storytelling," in *Proc. ACM ETRA*, Glasgow, UK, 2024. doi:10.1145/3649902.3653344.

[12] V. Mikhailova, **C. Gerhardt** *et al.*, "Age and Realism of Avatars in Simulated AR," in *Proc. IEEE VR*, Orlando, FL, 2024, pp. 83–93. doi:10.1109/VR58804.2024.00032.

[13] S. Arévalo Arboleda *et al.* incl. **C. Gerhardt**, "Beyond Looks: Agent Movement and Audiovisual Spatial Coherence in AR," in *Proc. IEEE VR*, Orlando, FL, 2024, pp. 502–512. doi:10.1109/VR58804.2024.00071.

[14] F. Weidner *et al.* incl. **C. Gerhardt**, "A Systematic Review on the Visualization of Avatars and Agents in AR & VR," *IEEE Trans. Vis. Comput. Graphics*, vol. 29, no. 5, pp. 2596–2606, 2023. doi:10.1109/TVCG.2023.324707

[15] **C. Gerhardt**, P. Aichroth, and S. Mann, "Selective Face Encryption in H.264 Encoded Videos," in *Proc. IEEE VCIP*, 2017, pp. 1–4. doi:10.1109/VCIP.2017.8305040.

Open-Source Projects & Datasets

AWARE: Weather-augmentation benchmark framework comprising three pipelines: **SWIFT** (augmentation), **PRISM** (generation), and **PROVE** (evaluation).
https://carhartt21.github.io/AWARE

AWACS: Weather-balanced augmented dataset — 71 400 images across 25 augmentation strategies for robustness evaluation.

SkyCloud360: 600 equirectangular panoramic images for sky and cloud segmentation.
Code: https://github.com/carhartt21/SkyCloud360 Data: https://osf.io/a5ews

SkyCloudNet: Sky and cloud segmentation network with pre-trained models.
Code: https://github.com/carhartt21/SkyCloudNet Data: https://osf.io/y69ah

OUTSIDE15k: 15 000 outdoor images with 24-class pixel-level annotations.
Code & Data: https://github.com/carhartt21/outsideNet

360° Detection: 2 500 panoramic images, 81 traffic sign classes.
Code: https://github.com/carhartt21/360-Object-Detection Data: https://osf.io/fsteu

Technical Skills

Languages: Python, C/C++/C#, Java, MATLAB/Octave, SQL/MySQL, LaTeX

ML / DL: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, ONNX

Tools: Git, Docker, Linux, Bash, Jupyter, Weights & Biases, Slurm

Domains: Semantic Segmentation, Object Detection, Image Augmentation, 360° Vision, VR/AR

Languages

German: Native

English: Fluent (C1+)

French: Basic knowledge

Serbian: Basic knowledge

Certifications

C-Lizenz (Coaching License) *2023*

EU Drone Pilot Certificate (A1/A3)

Funding

2019 – 2024: Research funded by the Free State of Thuringia (Grant No. 2017 FE 9147), co-financed by the European Union under the European Regional Development Fund (ERDF).

Academic Service

Teaching: Courses in VR/AR and computer vision at TU Ilmenau

Supervision: Co-supervision of Bachelor's and Master's theses

Ilmenau, March 13, 2026